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Earth Embeddings as Products

Taxonomy, Ecosystem, and Standardized Access

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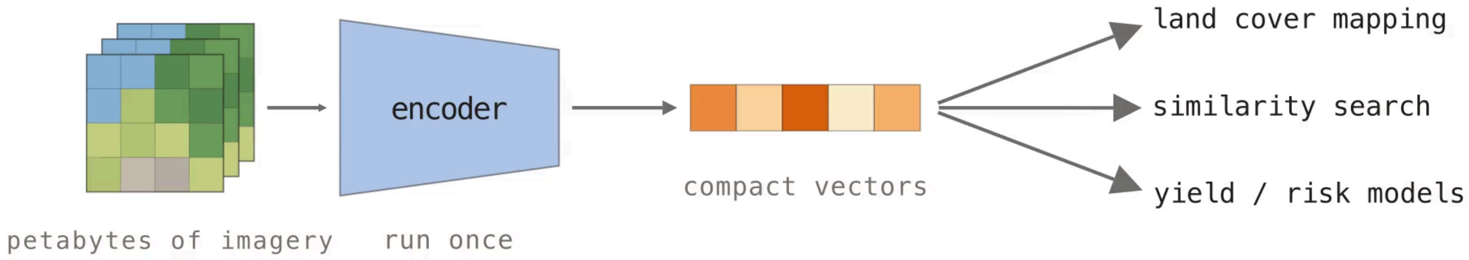
Embeddings as Reusable Data

NASA's EOSDIS alone holds **178.7 PB** of imagery and grows by 160 TB per day. Analyses of this archive each repeat the same download, preprocessing, and GPU inference. An embedding product runs the model once and distributes the output vectors as data.



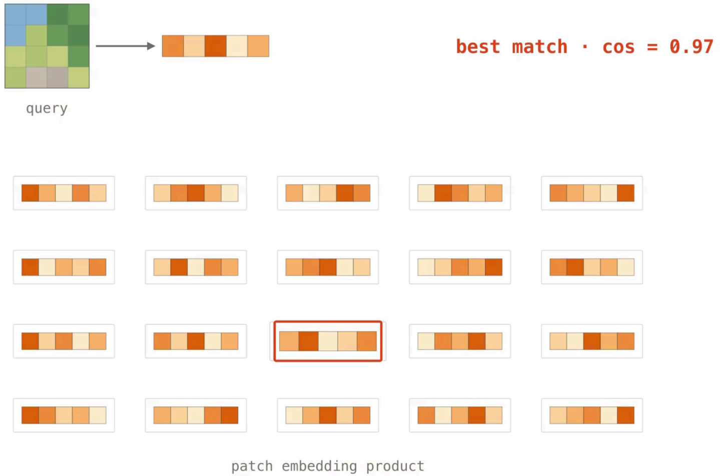
NASA ESDS annual metrics, FY2025 · Bommasani et al., 2021 — foundation model framing.

The Earth as One Large Document

Embedding retrieval matured in image search and in document retrieval for LLMs (RAG). The pipeline splits a corpus into chunks, embeds each chunk, and indexes the vectors for search.

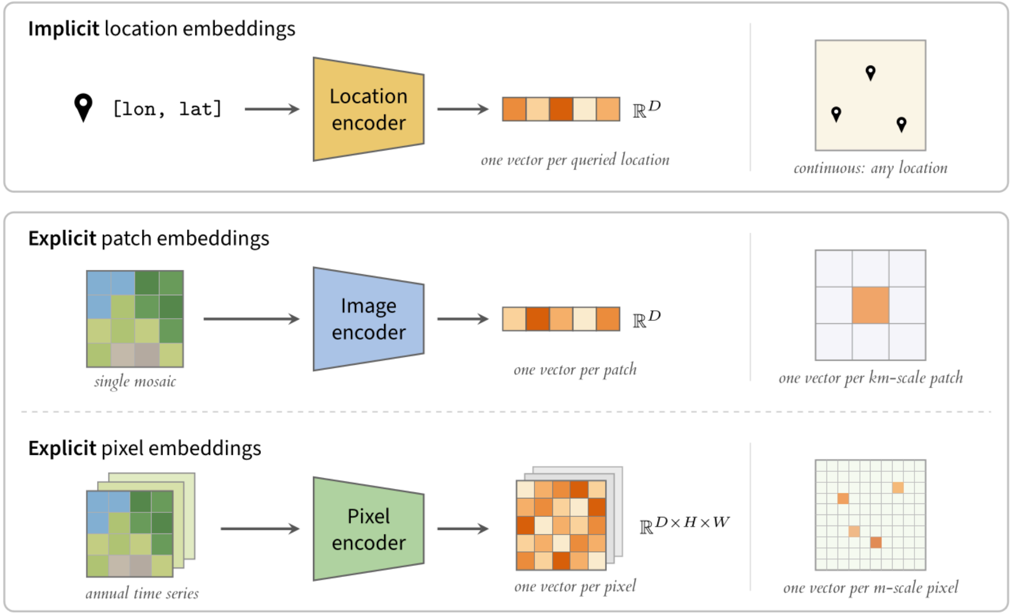
For the Earth archive, the chunk is not well defined:

What is the **spatial extent** of a chunk?	What is the **resolution** of each point in it?
One **timestamp**, or a time series?	Which **signal** (MSI band, SAR, elevation)?



Lewis et al., 2020 — retrieval-augmented generation · clip: query-by-example over a patch embedding product.

Three Families of Earth Embeddings



Stewart, Fang, Corley, Zhu — Earth Embeddings (book chapter, 2026), Fig. 1; figure design adapted from Klemmer et al., 2025 (EarthArXiv).

Patch Embedding Products

All known patch embedding products as of July 2026. *sparse spatial or temporal coverage.

Product	Extent	Resolution	Years	Dim.	Dtype
MOSAIKS (2021–25)	USA / Global	1 km / 0.01°	2018 / 2019	8192 / 4000	float32
Clay v0 / v1.5 (2024–26)	USA / Global	154 m – 5.12 km	2010–2025*	768–1024	float32
Major TOM (2024–26)	Global*	120 m – 3.84 km	2016–2024*	256–2048	float32
Earth Index (2025)	Global	320 m	2024	384	float32
Copernicus-Embed (2025)	Global	0.25°	2021	768	float32

Major TOM is the largest family, with more than a dozen model variants on one shared grid. All patch products store float32. One vector summarizes an entire mosaic tile, which suits retrieval more than dense mapping.

Earth Embeddings (book chapter, 2026), Table 2 — abridged to family level.

Pixel Embedding Products

All known pixel embedding products as of July 2026. Every product is built from annual time series. *sparse coverage.

Product	Extent	Resolution	Years	Dim.	Dtype
Presto Embeddings (2025)	Togo	10 m	2019–2020	128	uint16
Tessera Embeddings (2025)	Global*	10 m	2017–2025*	128	int8 → float32
Google Satellite Embedding (2025)	Global	10 m	2017–2025	64	int8 → float64
Embedded Seamless Data (2026)	Global	30 m	2000–2024	12	uint16 → float32

Google Satellite Embedding covers the full Sentinel era at 64 dimensions. **Embedded Seamless Data** trades resolution for 25 years of Landsat history. The int8 and uint16 dtypes reduce storage at these volumes.

Earth Embeddings (book chapter, 2026), Table 3.

Commercial Embedding Products and Services

Descartes Labs 2017

GeoVisual Search was the first planet-scale visual search over satellite imagery features. A query patch of a wind turbine returns wind turbines worldwide. It predates the current products by eight years.

Earth Genome 2025

Earth Index packages Clay embeddings on Source Cooperative into a search service for investigative journalists and conservation teams, who use it to find mining sites, airstrips, and feedlots.

LGND 2025

Raised **$9M** in 2025 to build services on geographic embeddings, and published the full global Clay v1.5 float32 corpus on Source Cooperative.

All three build their products on the embeddings rather than the model weights.

TechCrunch, Mar 2017 — GeoVisual Search · Ingold — Embeddings for All, Earth Genome, 2025 · LGND seed round, PR Newswire, Jul 2025.

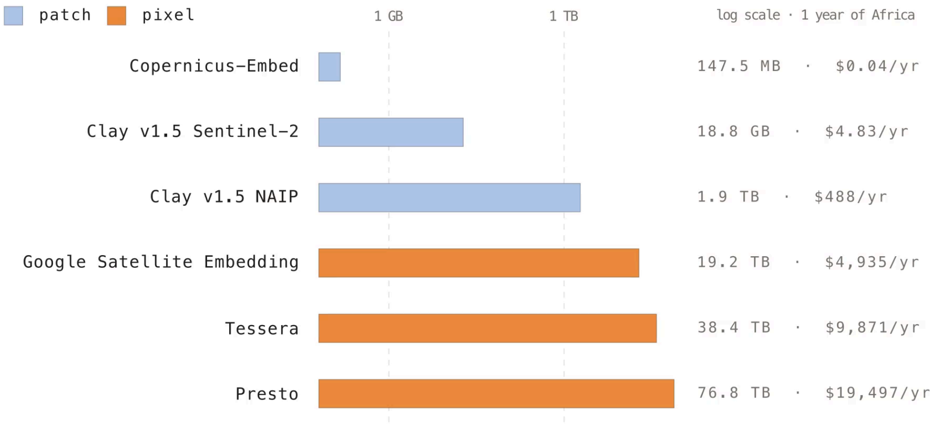
Fragmented Formats, Grids, and Hosting

- Products are scattered across **Source Cooperative** (Clay, Earth Index), Hugging Face (Major TOM), Earth Engine (Google, Presto), and private servers (Tessera).
- Formats span **GeoParquet**, GeoTIFF with an assumed CRS, and ungeoreferenced NumPy arrays.
- Each product defines its own **tiling grid**, so any cross-product comparison starts with reprojection.
- **One flipped coordinate** in the Google rasters forced fixes in **GDAL, rasterio, and TorchGeo**.

Each producer distributes differently, so integration effort is repeated for each product and each user.

Corley — The Technical Debt of Earth Embedding Products, cloudnativegeo.org, Feb 2026.

Storage and Egress Costs



Patch products stay in the MB–GB range, while dense 10 m pixel products reach tens of TB. A single full download of Presto's Africa coverage adds ~**$6.9k in egress** on top of storage.

Earth Embeddings (book chapter, 2026), Table 8 — AWS S3 Standard, us-west-2 · egress from Corley, 2026.

Hosting and Format Barriers

- **Hugging Face** enforces storage caps and API rate limits, so TB-scale downloads throttle or fail.
- **Tessera** distributes `.npy` tiles plus a metadata sidecar. NumPy arrays do not support HTTP range requests, so reads download whole tiles that a **COG or Zarr would stream**.
- The **AlphaEarth embeddings** were available only through Earth Engine or a requester-pays bucket. **Taylor Geospatial rehosted them in a non-requester-pays bucket on Source Cooperative**, with free egress, HTTP range reads, and CORS for browser streaming.
- Several products are so **sparse in space and time** that no common footprint exists for comparison.

Corley — The Technical Debt of Earth Embedding Products, cloudnativegeo.org, Feb 2026 · AEF rehost: source.coop, 2026.

Reproducibility and Licensing

- **Clay, Earth Index, and Copernicus-Embed** release code, weights, and data openly.
- **Tessera** releases code, weights, and embeddings openly, but records **no metadata about which inputs built each tile**, so its outputs cannot be audited.
- **Major TOM's** CC-BY-SA training data makes its embeddings copyleft, deterring commercial use.
- **AlphaEarth and ESDNet** publish CC-BY embeddings but keep code and weights proprietary.
- **No product provides checksums.** Reprocessed archives make the exact inputs unrecoverable.

Earth Embeddings (book chapter, 2026), Tables 4–6 — license provenance from data to weights to embeddings.

Foundation Models vs. Embedding Products

Foundation models

Distributed as public weights (DOFA, OlmoEarth). Users manage preprocessing and GPU inference themselves. This is flexible but raises a hardware and engineering barrier.

Embedding products

Distributed as pre-computed vector archives (Google Satellite Embedding). The assets are frozen and versioned, so analysis proceeds without the model or its compute.

A product is pinned to the data snapshot it was computed on, so **results measured on a product do not generalize to the model behind it**.

Fang, Stewart, Corley, Zhu, Azizpour — Earth Embeddings as Products, IGARSS 2026, §II.

Paper Metrics vs. Product Metrics

The AlphaEarth and Tessera papers report benchmark numbers from internal pipelines. The same tasks, evaluated on the **released annual embedding products**, give different numbers.

- The papers embed **exact input stacks**. The products are **annual composites** over reprocessed archives.
- Benchmarks should describe the **downloadable data** rather than the model.

Earth Embeddings (book chapter, 2026), §5 · Corley et al., 2026 — EuroSAT-Embed re-evaluation of AlphaEarth, Tessera, OlmoEarth.

No One Knows the SOTA in GFMs

An audit of 152 geospatial foundation model papers found **46 cross-paper disagreements of ≥ 10 points** for the same model, benchmark, and protocol.

94 of 126 papers use a pretraining configuration that appears in no other paper. **39% release no weights**, so their results cannot be re-run.

We built **torchgeo-bench**, a maintained harness for frozen backbones with shared datasets, consistent probes, and bootstrapped confidence intervals.


Taylor Geospatial

No One Knows the State of the Art in Geospatial Foundation Models

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github.com/taylor-geospatial/gfm-leaderboard

Abstract

Geospatial foundation models (GFMs) have been proposed as generalizable backbones for disaster response, land-cover mapping, food-security monitoring, and other high-stakes Earth-observation tasks. Yet the published work about these models does not give reviewers or users enough information to tell which model fits a given task. **We argue that nobody knows what the current state of the art is in geospatial foundation models.** The methods may be useful, but the GFM literature does not standardize evaluations, training and testing protocols, released weights, or pretraining controls well enough for anyone to compare or rank them. In a 152-paper audit, we find 46 cross-paper disagreements of at least 10 points for the same model, benchmark, and protocol; 94/126 papers with extractable pretraining data use a configuration no other paper uses; and 39% of GFM papers release no model weights. This lack of community standards can be solved. We propose six concrete expectations: named-license weight release, shared core evaluations, copied-versus-rerun baseline annotations, variance reporting, one shared evaluation harness, and data-vs-architecture-vs-algorithm controls. These gaps are a coordination failure, not a fault of any individual lab; the authors of this paper, like many others in the GFM community, have contributed to them. Rather than just critiquing the community, we aim to provide concrete steps toward a shared understanding of how to innovate GFMs.

1 Introduction

The promise of geospatial foundation models is cheap, easy reuse across domains. A single pretrained Earth-observation backbone should transfer across sensors, geographies, label regimes, and downstream tasks: crop mapping, flood mapping, building extraction, forest monitoring, land-cover change, and more. That promise makes evaluation harder than ordinary model comparison. A paper may compare M models across N benchmarks, but the benchmarks differ in spatial resolution, modality, class definitions, geographic coverage, label quality, and whether the reported metric measures accuracy on small image patches or the quality of an actual map a user would rely on. This mirrors the benchmark-lottery problem described in broader ML evaluation work [16]: benchmark choice can dominate apparent progress when communities lack shared protocols. The GFM community therefore needs *clearer standards on how to test and compare GFMs*.

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Preprint.

Fusing Multiple Products

You do not have to pick one product.

Fusing AlphaEarth, Tessera, GeoCLIP, and SatCLIP beats the best single product on **4 of 6 downstream tasks**.

Which products complement each other depends on the task and region, so a probe trained on the concatenated embeddings is a cheap first experiment.


BETTER TOGETHER: EVALUATING THE COMPLEMENTARITY OF EARTH EMBEDDING MODELS

A PREPRINT

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ABSTRACT

Earth embedding models transform Earth observation data into embeddings uniquely tied to locations on the Earth's surface. These models are typically evaluated in isolation, comparing the downstream task performance across different Earth embeddings. However, spatially aligned embeddings can naturally be fused, providing richer information per location, a capability that isolated evaluations fail to capture. We therefore propose assessing Earth embeddings by their *complementarity*: the performance gain of fused embeddings over the best single-model baseline. To operationalise this, we introduce an embedding complementarity index applicable to any embedding and task, and evaluate four Earth embedding models (AlphaEarth, Tessera, GeoCLIP, SatCLIP) in isolation, in all pairs, and jointly across six downstream tasks. Fused embeddings outperform the best single model in four out of six tasks, confirming that single-embedding evaluations often underestimate Earth embedding capabilities. Complementarity proves both task- and location-dependent. Further, for a land cover regression task, we find that complementarity is partially determined by the spatial scale of land cover classes. Complementarity reframes Earth embeddings: the greatest future gains may come not from any single Earth embedding model, but from combinations that are better together.

1 Introduction

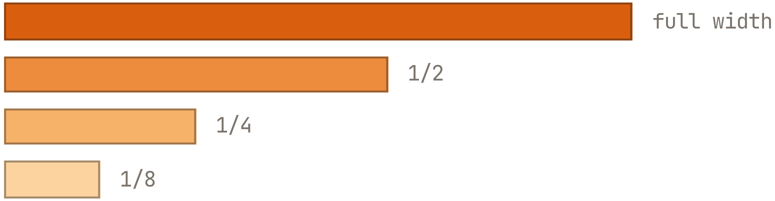
Earth embedding models construct compressed embeddings from rich, multimodal Earth observation data, which can be used for a variety of geospatial downstream tasks [Klemmer et al., 2025b]. Compared to foundation model embeddings from other domains, such as vision or language, Earth embeddings have the unique property that any embedding is inherently linked to a location on the Earth's surface. This trait has turned Earth embeddings into commodities, which are pre-computed once and then retrieved by coordinates alone, bypassing the need for model inference [Fang et al., 2026]. As a result of this ease of access, pre-computed Earth embeddings have now been used for a range of downstream tasks in agriculture, environmental sciences and ecology [Zvonkov et al., 2025, Ma et al., 2026, Young et al., 2026, Lisaius et al., 2026, Ball et al., 2026]. But while the *differences* in task performance between Earth embeddings are regularly assessed [Zvonkov et al., 2025, Lisaius et al., 2026, Ball et al., 2026], the *complementarity* between different Earth embeddings is not.

This lack of complementarity assessments might stem from the notion that Earth embedding models are expected to encode all possible information about a location within a single vector. Yet, Earth embedding models are developed with varying training objectives, input data modalities and training datasets. For example, explicit geospatial foundation models map Earth observation images to pixels embeddings, and are trained using masked auto-encoders [Brown et al., 2025] or pixel-wise contrastive loss [Tseng et al., 2023, Feng et al., 2025], while implicit neural representation models map coordinates to location embeddings via various contrastive loss objectives [Vivanco et al., 2023, Klemmer et al., 2025a, Dhakal et al., 2025]. This variety of Earth embeddings suggests that each might capture a different "view" of the same location that is partially unique, and consequently they can be complementary sources of information. To test this,

van der Plas et al., 2026 — Better Together, arXiv:2605.18667.

Open Research Directions

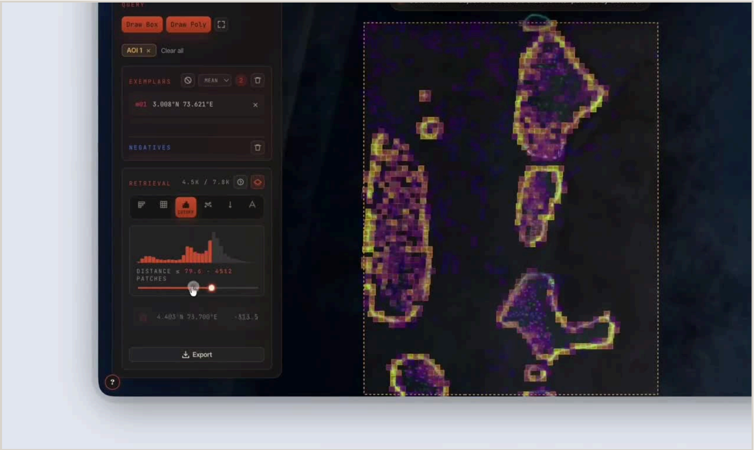
- **Quantization.** Google and Tessera use int8 at no measurable cost. Binary recovers ~65% of float32 nearest neighbors.
- **Disentangled representations.** VAE-style training gives each dimension a separate meaning. Not yet applied to Earth embeddings.
- **Matryoshka learning.** Tessera v2, Clay, and our **MIND location encoder** train nested dimensions, so users truncate to their budget.



the leading dimensions form a usable embedding at any width

Kusupati et al., 2022 — Matryoshka representation learning · Corley & Robinson, 2026 — Compressing Earth Embeddings · MIND: Corley et al., 2026.

Compressed Embeddings in the Browser



TerraBit — 50M Clay v1.5 patches binarized to 128 bytes each and streamed from Source Cooperative. Hamming search runs in a Web Worker with no backend.

isaac.earth/terrabit

DeltaBit — AlphaEarth pixel differences at 8 bytes per pixel (PCA-8 + int8) as XYZ GeoTIFF tiles. The user labels, trains, and maps change in the browser (Seattle, 2020 → 2024).

calebrob.com/deltabit

Recommendations for Producers

Embedding type	Format
Location / implicit	ONNX or PyTorch
Patch / region	GeoParquet
Pixel, snapshot	GeoTIFF or GeoZarr
Pixel, time series	GeoZarr

- Provide a **model card:** sensors, time window, CRS, grid, dtype, license.
- Store metadata **inside the files** rather than in separate docs.
- **int8 by default**; PCA to 64 dims costs <2% accuracy for 64× compression.
- **Runnable benchmarks** instead of leaderboards.

Earth Embeddings (book chapter, 2026), §7 · Corley & Robinson, 2026 — TerraBit compression study.

Takeaways and Open Problems

Pick the family by task: implicit for location context, patch for retrieval, pixel for dense mapping. Validate under geographic splits against simple baselines, and consider fusing products instead of picking one. Open problems are standardized formats and provenance, ocean and atmosphere coverage, uncertainty layers, and shared benchmarks (identical models differ by more than ten points across papers).

Chapter arxiv.org/abs/2608.03410

Survey github.com/hfangcat/Awesome-Geospatial-Embeddings

Bench torchgeo.org/torchgeo-bench

Contact isaac.corley@taylorgeospatial.org

these slides